The Lagrangian Subspace Noise Problem: A New Framework for Post-Quantum Cryptography

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Abstract

We introduce the Lagrangian Subspace Noise (LSN) problem — a variant of the Learning Parity with Noise (LPN) problem where the secret is a Lagrangian subspace of a symplectic vector space rather than a linear function. The secret space, the Lagrangian Grassmannian $\mathrm{Lagr}(2n, \mathbb{F}_2)$, has cardinality $2^{n(n+1)/2+O(1)}$, exponentially larger than LPN's 2^n yet tightly constrained by the symplectic form. We prove an unconditional exponential Statistical Query (SQ) lower bound of $\Omega(2^n)$ queries via an explicit symplectic spread, and a conditional $2^{2n-O(1)}$ bound under a precisely-stated extremality conjecture, together with an exact pairwise-correlation formula and no asymptotic error term. We establish that linear and low-degree polynomial feature-map reductions are information-theoretically impossible, that adaptive linear SQ algorithms have only exponentially small advantage, and we map the exact standard-model gap that remains for general non-linear reductions. We construct two cryptographic primitives from LSN: a succinct non-interactive argument (LSN-SNARK) with $O(n^2)$ R1CS constraints ($\approx$ 4,225 for $n = 65$, compared to millions for ZK-lattice proofs), and a key-encapsulation mechanism (LSN-KEM) with concatenated polar-code reconciliation achieving 80-bit security with public keys of 1.78 KB and ciphertexts of 288 B. All security claims are explicitly classified as theorem, evidence, or conjecture, and the remaining open problems are stated precisely.

Keywords. Post-quantum cryptography, Learning Parity with Noise, statistical query model, symplectic geometry, polar codes, zero-knowledge proofs.

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Data and code availability. All Python prototypes, test vectors, and the LaTeX source are available at `https://github.com/Gattochoo/lsn-pq`.

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1 Introduction

1.1 Background

The Learning Parity with Noise (LPN) problem, introduced by Blum *et al.* [BFKL94], asks to recover a secret linear function $s \in \mathbb{F}_2^k$ given noisy inner products $(a_i, \langle a_i, s\rangle \oplus e_i)$. Despite decades of cryptanalytic effort, no polynomial-time algorithm is known for LPN with constant noise rate. The best classical attacks run in sub-exponential time $2^{O(n/\log n)}$ using the Blum–Kalai–Wasserman (BKW) algorithm [BKW03] and its modern refinements [LF06, Lyu05, EKM17, GJL14, ZJW16, WS21]. Grover's algorithm reduces the naive key-search cost from 2^k to $2^{k/2}$ [Gro96], and quantum variants of BKW remain the dominant quantum threat; no published exponential quantum speed-up over the best classical methods is known [Kir11].

LPN has been the foundation for a wide range of cryptographic primitives, including OWFs, PKE, OT, and MPC [ABW10, Ale11, DGM14, BCGI18]. Its main theoretical limitation is the absence of a worst-case hardness reduction comparable to Regev's LWE reduction from lattice problems [Reg05]. LPN hardness is therefore treated as a *primitive assumption*, accepted on the basis of sustained cryptanalytic resistance rather than reducibility.

Post-quantum cryptography currently rests on six classical hardness families: multivariate quadratic equations (MQ), lattices (LWE/SIS), error-correcting codes (LPN/syndrome decoding), hash functions, Ising models, and tensor problems. A seventh family — genuinely new, structurally distinct, and not reducible to the existing six — has been sought since the inception of the field. LSN is the strongest current candidate for such a family.

1.2 The LSN Problem

We propose *Lagrangian Subspace Noise* (LSN), also referred to as *Learning Stabilizers with Noise* or *symplectic LPN*. The secret is not a linear function but a Lagrangian subspace $L \subset \mathbb{F}_2^{2n}$ of a symplectic vector space. The learner receives samples

$$(a_i, b_i) \quad \text{where} \quad b_i = \mathbf{1}_L(a_i) \oplus e_i,$$

with $a_i \sim \mathbb{F}_2^{2n}$ uniform, $e_i \sim \mathrm{Bernoulli}(p)$, and $\mathbf{1}_L$ the indicator of L. The secret space is the Lagrangian Grassmannian:

$$|\operatorname{Lagr}(2n, \mathbb{F}_2)| = \prod_{i=1}^{n}(2^i + 1) = 2^{n(n+1)/2+O(1)}. \tag{1}$$

This is exponentially larger than LPN's 2^n secrets, yet it is not an unstructured space: every secret satisfies the strong isotropy constraint $L = L^{\perp_\Omega}$. It is exactly this tension — exponential size versus rigid structure — that creates both the hardness and the technical difficulty of LSN.

Recent independent work by Khesin *et al.* [KLP+25], Poremba–Quek–Shor [PQS26], and Lu *et al.* [LPQR26] has shown that LSN is at least as hard as constant-rate LPN even for a single logical qubit, and has established strong barriers against reducing symplectic LPN to standard LPN (for linear reductions, proven in the constant-rate sample regime $m = \Theta(n)$; see Section 9). These results position LSN as the strongest current candidate for a genuinely new post-quantum hardness family outside the six classical ones (multivariate-quadratic, lattice, code, hash, Ising, tensor).

1.3 Contributions

Our contributions are organized around proof, barriers, and constructions.

1. **Exact SQ lower bound (Section 6).** We prove an unconditional $\Omega(2^n)$ SQ query lower bound via an explicit symplectic spread, and a conditional $q \geq 2^{2n-O(1)}$ bound under a pencil-extremality conjecture. Both rest on an exact pairwise-correlation formula

$$\langle D_L, D_{L'} \rangle = \frac{(1-2p)^2}{p(1-p)} \cdot 2^{j-2n},$$

where $j = \dim(L \cap L')$ and $\langle D_L, D_{L'} \rangle$ denotes the pairwise correlation. For $p = 1/4$ the coefficient is exactly $4/3$. A Markov concentration argument shows that a subset of size 2^{2n} with average correlation $O(2^{-2n})$ exists; the stronger worst-case bound remains open. The results hold for adaptive SQ algorithms.

2. **Reduction barriers (Section 9).** We prove that linear feature maps give zero advantage against LSN; polynomial feature maps of degree $< n$ have error at least 2^{-n}; and adaptive linear SQ algorithms have advantage at most $(1-2p)2^{-n}$ at any noise rate p. Together these barriers eliminate the natural reduction routes and localize the remaining open question to genuinely non-linear, adaptive reductions.

3. **Cryptographic primitives (Section 8).**
   - **LSN-SNARK:** A preprocessing SNARK for LSN membership with $O(n^2)$ R1CS constraints via the Siegel-chart (graph-Lagrangian) representation. Verified counts: 64 for $n = 8$, 1,764 for $n = 42$, 4,225 for $n = 65$.
   - **LSN-KEM:** A CCA-secure KEM via concatenated polar codes. Direct polar coding at $p = 1/4$ fails ($Z_0 = 0.866$, SC bound $P_e > 1$), so we concatenate an inner repetition code with an outer polar code. For $r = 7$ repetition we obtain effective noise $p' = 0.0706$, SC bound $P_e < 2^{-81}$, and parameters PK = 1.78 KB, CT = 288 B at 80-bit security. For $r = 11$ we obtain $P_e < 2^{-149}$ at 128-bit security.

4. **Rigorous caveats (Section 10).** Every claim is classified as theorem, evidence, or conjecture. We state explicitly the standard-model gap — the absence of a non-vacuous LPN(low-noise) $\to$ LSN or LWE $\to$ LSN reduction — that must close for full community acceptance.

2 Preliminaries

2.1 Symplectic Vector Spaces

Let $V = \mathbb{F}_2^{2n}$ with standard symplectic form

$$\Omega(x, y) = \sum_{i=1}^{n} (x_{2i-1}y_{2i} + x_{2i}y_{2i-1}).$$

A subspace $L \subset V$ is *isotropic* if $\Omega|_{L\times L} = 0$, and *Lagrangian* if it is isotropic of dimension n; equivalently $L = L^{\perp_\Omega}$. We write $\mathrm{Lagr}(2n, \mathbb{F}_2)$ for the Lagrangian Grassmannian.

Proposition 2.1 (Lagrangian count [Mac71])**.** $|\mathrm{Lagr}(2n, \mathbb{F}_2)| = \prod_{i=1}^{n}(2^i + 1) = 2^{n(n+1)/2+O(1)}$.

The intersection of two Lagrangians has even codimension. For $L, L' \in \mathrm{Lagr}(2n)$ let $j = \dim(L \cap L')$. The distribution of j is governed by the q-binomial coefficients at $q = 2$:

$$\Pr[j = k] = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_2 \cdot 2^{(n-k)(n-k+1)/2}}{|\mathrm{Lagr}(2n, \mathbb{F}_2)|}. \tag{2}$$

2.2 Fourier Transform on the Symplectic Space

The unnormalized Ω-Fourier transform of $f\colon \mathbb{F}_2^{2n} \to \mathbb{R}$ is

$$\mathcal{F}_\Omega[f](\xi) = \sum_{x \in \mathbb{F}_2^{2n}} f(x)(-1)^{\Omega(x,\xi)}.$$

Lemma 2.2 (Self-duality)**.** *For every* $L \in \mathrm{Lagr}(2n, \mathbb{F}_2)$*,*

$$\mathcal{F}_\Omega[\mathbf{1}_L] = 2^n \cdot \mathbf{1}_L.$$

Proof. If $\xi \in L$, then $\Omega(x, \xi) = 0$ for all $x \in L$, so the sum is $|L| = 2^n$. If $\xi \notin L$, choose $x_0 \in L$ with $\Omega(x_0, \xi) = 1$; the involution $x \mapsto x + x_0$ pairs terms of opposite sign, so the sum is 0. $\square$

2.3 Statistical Query Model

Let $\mathcal{D}$ be a class of distributions and D_0 a reference distribution. The *statistical dimension with average correlation* [FGR+17] is

$$\mathrm{SDA}(B(\mathcal{D}, D_0), \gamma) = \max\Big\{d : \forall S \subseteq \mathcal{D}, |S| \geq \frac{|\mathcal{D}|}{d} \Rightarrow \frac{1}{|S|^2} \sum_{D,D' \in S} |\langle D, D' \rangle| \leq \gamma\Big\}.$$

Theorem 2.3 (Feldman *et al.* [FGR+17])**.** *If* $\mathrm{SDA}(B(\mathcal{D}, D_0), \gamma) = d$*, any SQ algorithm distinguishing* $\mathcal{D}$ *from* D_0 *with success probability* $\alpha > 1/2$ *requires* $q \geq (2\alpha - 1)d$ *queries to* $\mathrm{VSTAT}(1/(3\gamma))$*.*

The theorem applies to randomized *adaptive* SQ algorithms, so our lower bound will inherit adaptivity automatically.

2.4 Notation

We write D_L for the LSN distribution with secret L and noise rate p, and D_0 for the noise-only distribution in which $b \sim \mathrm{Bernoulli}(p)$ independently of a. All logarithms are base 2 unless stated otherwise.

3 Related Work

LPN foundations and cryptanalysis. LPN was introduced by Blum, Furst, Kearns, and Lipton [BFKL94] as a foundation for cryptographic primitives. The dominant classical attack is BKW [BKW03], later optimized with fast Walsh–Hadamard transforms [LF06], covering codes [GJL14, ZJW16], and hybrid information-set-decoding techniques [EKM17, WS21]. Lyubashevsky [Lyu05] gave a polynomial-sample variant, and a comprehensive unified attack framework was developed by Esser, Kübler, and May [EKM17]. Quantum speed-ups are limited to Grover [Gro96] and quantum BKW [Kir11]; no published exponential improvement is known.

LPN variants. Many structured variants have been proposed to improve efficiency: Ring-LPN [HKL+12], LPN with structured noise [DP12], and the Learning with Rounding (LWR) problem [BPR12]. All of these retain a secret space of size $2^{O(n)}$. LSN differs fundamentally in that its secret space has size $2^{n^2/2+O(n)}$ while remaining polynomial-time sampleable via symplectic linear algebra.

Symplectic and quantum coding. Stabilizer codes, the quantum analogue of classical linear codes, are naturally described by Lagrangian subspaces of $\mathbb{F}_2^{2n}$ [Got97, CRSS98, NC00]. Quantum state tomography of stabilizer states under depolarizing noise is a closely related learning problem [GL22]. The recent independent work of Khesin *et al.* [KLP+25] proved that constant-rate LPN reduces to LSN even for $k = 1$, while Lu *et al.* [LPQR26] proved that linear reductions cannot reduce sympLPN to LPN in the constant-rate sample regime $m = \Theta(n)$; for $m = \omega(n)$ their bound shows error weight $1/2 - \delta$, which they note is not sufficient to fully rule out decodability (though they expect the barrier to extend to all $m = \mathrm{poly}(n)$).

Statistical query hardness. The SQ model was introduced by Kearns [Kea98] and developed by Blum *et al.* [BFJ+94]. Feldman, Grigorescu, Reyzin, and Vempala [FGR+17] gave the general SDA framework we use, and Diakonikolas *et al.* [DKS17] applied it to high-dimensional robust estimation. Quantum algorithms that access only classical samples can achieve at most a quadratic Grover speed-up over classical algorithms. Since the SQ lower bound is exponential in n, such a speed-up does not affect the asymptotic security parameter, making SQ lower bounds relevant evidence for quantum hardness.

Code-based cryptography and polar codes. Code-based cryptography dates to McEliece [McE78] and Prange [Pra62]. Modern standards include Classic McEliece [ABD+21], HQC [AAC+22], and BIKE. Polar codes, introduced by Arikan [Ari09], are capacity-achieving with explicit construction; list decoding was introduced by Tal and Vardy [TV15], and finite-length analysis was developed by Mori–Tanaka [MT09], Trifonov [Tri12], and Hassani *et al.* [HVU14]. We use Bhattacharyya bounds for conservative parameter selection [Sei15].

Post-quantum KEMs. The NIST PQC standardization process selected Kyber (lattice) [SAB+22], Classic McEliece, and HQC [AAC+22] as KEM candidates. LSN-KEM offers competitive sizes with a fundamentally different assumption.

4 The LSN Problem

Definition 4.1 ($\mathrm{LSN}_{n,m,p}$)**.** Let $L \subset \mathbb{F}_2^{2n}$ be uniformly random Lagrangian. The LSN distribution D_L over $\mathbb{F}_2^{2n} \times \mathbb{F}_2$ is generated by sampling $a \sim \mathbb{F}_2^{2n}$, $e \sim \mathrm{Bernoulli}(p)$, and outputting (a, b) where $b = \mathbf{1}_L(a) \oplus e$. Given m independent samples from D_L, recover L.

Search vs. decision. All hardness results in this paper are stated for the *decisional* problem: distinguishing D_L from the noise-only distribution D_0 (where $b \sim \mathrm{Bernoulli}(p)$ independently of a). The search-to-decision reduction follows the same template as for LPN [BFKL94]: given a decision oracle and a candidate Lagrangian L', one can verify whether $L' = L$ by testing whether the samples are consistent with $D_{L'}$; a standard hybrid argument then yields search hardness from decision hardness. We therefore treat decisional LSN as the primitive assumption.

4.1 Distance Distribution and Expected Intersection

The correlation between two LSN distributions depends only on the dimension $j = \dim(L \cap L')$ of the intersection of their secrets. The distribution of j is:

Theorem 4.2 (Distance distribution)**.** *For uniform $L, L' \in \mathrm{Lagr}(2n, \mathbb{F}_2)$,*

$$\Pr[j = k] = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_2 \cdot 2^{(n-k)(n-k+1)/2}}{|\,\mathrm{Lagr}(2n, \mathbb{F}_2)|}.$$

Numerical evaluation shows that $\mathbb{E}[j] \to 0.76$ and $\mathrm{Var}(j) \to 0.60$ as $n \to \infty$.

The rapid decay of Equation (2) means that two random Lagrangians typically intersect in dimension 0 or 1; intersections of dimension ≥ 14 occur with probability $< 2^{-100}$ for $n \geq 42$.

4.2 Parameters

Table 1 gives the exact security parameters derived in Section 6.

Table 1: LSN security parameters for $p = 1/4$.

Security	n	$\log_2(q_{\min})$[a]	$\lvert \mathrm{Lagr} \rvert$	PK (KEM)
80-bit	41	80.6	$\sim 2^{861}$	1.78 KB
128-bit	65	128.6	$\sim 2^{2145}$	2.78 KB
192-bit	97	192.6	$\sim 2^{4753}$	—
256-bit	129	256.6	$\sim 2^{8385}$	—

[a] The $q_{\min}$ figures use the conditional bound of Theorem 6.6; the unconditional spread bound (Theorem 6.4) gives $\Omega(2^n)$.

The $q_{\min}$ figures use the exact pairwise-correlation formula of Lemma 6.1, so the 80-bit line is met *exactly* at $n = 41$ with no asymptotic slack. The constant factor in the exponent is ≈ 0.67 (below 1), so parameters with $2n < \lambda$ are not advised without a rigorous upper bound. PK sizes are shown only for the two KEM parameter sets derived in Section 8.2; 192/256-bit KEM parameters depend on the inner repetition length r and are left as standard extrapolations.

5 Decoder Landscape

We review the natural decoder families and explain why each fails against LSN at constant noise.

Linear decoders. A real-linear query has the form $q(a, b) = \langle w, a \rangle + c \cdot b$. Its expectation under D_L is

$$\mathbb{E}_{D_L}[q] = c \cdot \Pr[b = 1] = c \cdot \big(p + 2^{-n}(1 - 2p)\big),$$

which is independent of L. Hence real-linear decoders have exactly zero L-identification power. $\mathbb{F}_2$-linear queries can achieve advantage at most $(1 - 2p)2^{-n}$ (Theorem 6.8).

Polynomial decoders. The indicator $\mathbf{1}_L$ has an exact degree-n polynomial representation

$$\mathbf{1}_L(x) = \prod_{j=1}^{n} (1 + \langle w_j, x \rangle)$$

where $\{w_j\}$ is a basis for the annihilator of L. Any polynomial of degree $< n$ misses at least one factor and therefore has error $\geq 2^{-n}$.

List decoding. The list size needed to enumerate all candidate Lagrangians is $\lvert \mathrm{Lagr} \rvert = 2^{n^2/2+O(n)}$. Even with pruning, list decoding is infeasible for $n \geq 41$.

BKW and ISD. BKW would require treating the secret as an n^2-bit object, giving complexity $2^{\Omega(n^2/\log n)}$. Information-set decoding does not apply directly because the code ensemble (Lagrangian indicators) is highly structured rather than random linear.

Structural attacks. For very low noise ($p = o(1/n)$), the span of the positive samples reveals L. At constant noise $p = 1/4$, [LPQR26] and our experiments show that structural attacks are ineffective: the span of positives has full dimension after $O(n)$ samples.

Empirical sample-complexity lower bound. We ran brute-force experiments for $n = 3, 4, 5$ using transvection-orbit enumeration of the full Lagrangian family. In each trial we sampled $L \leftarrow$ Lagr uniformly, drew m noisy samples from $\mathsf{LSN}_{L,1/4}$, and let a brute-force decoder enumerate all Lagrangians, picking the one with the highest agreement. The recovery success rates (200 trials for $n = 3$, 50 for $n = 4$, 20 for $n = 5$) were:

n	$\lvert$ Lagr $\rvert$	$m = 2^{2n-2}$	$m = 2^{2n-1}$	$m = 2^{2n}$	$m = 2 \cdot 2^{2n}$
3	135	11.5%	18.5%	46.5%	80.0%
4	2,295	6.0%	24.0%	62.0%	98.0%
5	75,735	5.0%	20.0%	75.0%	100% (20/20)

Two trends are visible. First, the success rate at the floor $m = 2^{2n}$ *increases* with n (46.5%, 62%, 75%), consistent with 2^{2n} being the asymptotic threshold (the constant factor in front appears to be below 1 and to matter less as n grows). Second, halving the budget to $m = 2^{2n-1}$ already drops recovery below 25% even at $n = 5$, so the threshold sits at the 2^{2n} scale rather than a constant factor below it. These experiments *empirically support* a 2^{2n} sample-complexity scale for the brute-force ML decoder, matching the pairwise floor of Lemma 6.1; they do not by themselves rule out a more sample-efficient decoder—the rigorous lower bound is the SQ query bound of Section 6.

6 Statistical Query Lower Bound

This section contains the main technical result: an exponential SQ lower bound with an exact correlation formula.

6.1 Exact Pairwise Correlation

Let D_0 be the noise-only distribution. The likelihood ratio of D_L with respect to D_0 is

$$\ell_L(a, b) = \frac{dD_L}{dD_0}(a, b) - 1 = \mathbf{1}_L(a) \cdot \beta \cdot \Big(\mathbf{1}_{b=1} - \mathbf{1}_{b=0}\frac{p}{1-p}\Big),$$

where $\beta = (1 - 2p)/p$.

Lemma 6.1 (Exact pairwise correlation)**.** *For $L, L' \in \mathrm{Lagr}(2n, \mathbb{F}_2)$ with $j = \dim(L \cap L')$,*

$$\langle D_L, D_{L'}\rangle = \frac{(1-2p)^2}{p(1-p)} \cdot 2^{j-2n}. \tag{3}$$

Proof. Because a and b are independent under D_0, the inner product factors:

$$\langle D_L, D_{L'}\rangle = \mathbb{E}_a[\mathbf{1}_L(a)\mathbf{1}_{L'}(a)] \cdot \mathbb{E}_b\Big[\beta^2\Big(\mathbf{1}_{b=1} - \mathbf{1}_{b=0}\frac{p}{1-p}\Big)^2\Big].$$

The first factor is $|L \cap L'|/2^{2n} = 2^{j-2n}$. The second factor is

$$\beta^2\Big(p + (1-p)\frac{p^2}{(1-p)^2}\Big) = \frac{(1-2p)^2}{p^2} \cdot \frac{p}{1-p} = \frac{(1-2p)^2}{p(1-p)}.$$

Multiplying yields Equation (3). For $p = 1/4$ the coefficient is exactly 4/3. □

6.2 Average Correlation

Taking expectation over j gives the average pairwise correlation.

Lemma 6.2 (Average correlation)**.** *Let $C_n = \mathbb{E}[2^j]$ computed from Theorem 4.2. Then $C_n \to 2$ and*

$$\rho_{\mathrm{avg}} = \frac{(1-2p)^2}{p(1-p)} \cdot C_n \cdot 2^{-2n}.$$

6.3 SDA Concentration

Lemma 6.3 (Existence of low-correlation subset)**.** *Let $\gamma = 2\rho_{\text{avg}}$. There exists a subset $\mathcal{D}' \subset \{D_L\}$ of size $|\mathcal{D}'| = 2^{2n}$ with average correlation at most γ.*

Proof. Consider a uniformly random subset $S \subset \text{Lagr}(2n)$ of size $M = 2^{2n}$. By symmetry of the Lagrangian Grassmannian under $\text{Sp}(2n)$, $\mathbb{E}[\rho(S)] = \rho_{\text{avg}}$. Markov's inequality gives

$$\Pr[\rho(S) > 2\rho_{\text{avg}}] < \frac{\mathbb{E}[\rho(S)]}{2\rho_{\text{avg}}} = \frac{1}{2}.$$

Hence with probability at least $1/2$, a random subset of size 2^{2n} has average correlation at most $\gamma = 2\rho_{\text{avg}}$. This proves existence of such a subset. A worst-case bound over *all* subsets of size 2^{2n} would require a structural argument; we discuss this in Section 10. $\square$

6.4 Unconditional SQ Bound: The Symplectic Spread

We first give an unconditional exponential lower bound on a concrete worst-case promise problem. The bound is weaker in both query count and VSTAT strength than the conditional average-case bound of Theorem 6.6, but it requires no unproven structural hypothesis.

Let $V = \mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ with symplectic form $\omega((a,b),(c,d)) = \text{Tr}(ad) + \text{Tr}(bc)$, where Tr is the trace from $\mathbb{F}_{2^n}$ to $\mathbb{F}_2$. The *Desarguesian symplectic spread* $\mathcal{S}^* = \{L_\lambda\}_{\lambda \in \mathbb{F}_{2^n}} \cup \{L_\infty\}$ consists of $d_0 = 2^n + 1$ Lagrangians

$$L_\lambda = \{(x, \lambda x) : x \in \mathbb{F}_{2^n}\}, \qquad L_\infty = \{0\} \times \mathbb{F}_{2^n},$$

which are pairwise transversal: $\dim(L \cap L') = 0$ for all distinct $L, L' \in \mathcal{S}^*$.

Theorem 6.4 (Unconditional spread bound)**.** *Let $\kappa = (1-2p)^2/(p(1-p))$ and fix $1 \le t \le n-1$. Set $\bar{\gamma}_t = \kappa \cdot 2^{-(2n-t)}$. Every subset $T \subseteq \mathcal{S}^*$ with $|T| \ge 2^{n-t}$ has diagonal-inclusive average correlation at most $2\bar{\gamma}_t$. Consequently $\text{SDA}(\mathcal{S}^*, 2\bar{\gamma}_t) \ge 2^t$, and any SQ algorithm that is correct on every $L \in \mathcal{S}^*$ requires $\Omega(2^t)$ queries to $\text{VSTAT}(1/(6\bar{\gamma}_t))$.*

At $t = n-1$ this yields an unconditional $\Omega(2^n)$ query lower bound at VSTAT strength $O(2^n)$.

Proof. For distinct $L, L' \in \mathcal{S}^*$ we have $j = \dim(L \cap L') = 0$, so by Lemma 6.1 the pairwise correlation is exactly $\langle D_L, D_{L'} \rangle = \kappa \cdot 2^{-2n} = \bar{\gamma}_t \cdot 2^{-t} \le \bar{\gamma}_t$. The diagonal terms contribute $\beta = \kappa \cdot 2^{-n}$ each, so for $|T| \ge 2^{n-t}$

$$\frac{1}{|T|^2} \sum_{D,D' \in T} |\langle D, D' \rangle| \;\le\; \frac{|T|^2 \cdot \bar{\gamma}_t + |T| \cdot \beta}{|T|^2} \;\le\; \bar{\gamma}_t + \frac{\beta}{2^{n-t}} \;\le\; 2\bar{\gamma}_t.$$

Hence $\text{SDA}(\mathcal{S}^*, 2\bar{\gamma}_t) \ge |\mathcal{S}^*|/2^{n-t} \ge 2^t$. Applying Theorem 2.3 with $\alpha = 2/3$ gives the query bound. $\square$

Honesty notes. (i) **Worst-case promise only.** The spread has measure $|\mathcal{S}^*|/|\text{Lagr}(2n, \mathbb{F}_2)| \sim 2^n/2^{n(n+1)/2} \to 0$, so Theorem 6.4 does *not* cover the average-case (uniform L) problem; it protects only the promise problem "$L \in \mathcal{S}^*$?". (ii) **VSTAT trade-off.** The unconditional bound is $\Omega(2^n)$ queries at $\text{VSTAT}(O(2^n))$; the conditional bound below is $2^{2n-O(1)}$ queries at $\text{VSTAT}(\Theta(2^{2n}))$. The two parameters are not interchangeable. (iii) The subfamily restriction is valid for the promise problem: any algorithm solving LSN over the full family is correct on $\mathcal{S}^*$ a fortiori.

6.5 Conditional SQ Bound

The stronger 2^{2n}-scale bound rests on a structural conjecture about extremal subsets.

Conjecture 6.5 (Pencil extremality)**.** *There is an absolute constant c such that every subset $\mathcal{D}' \subseteq \{D_L\}$ of size $|\mathcal{D}'| \geq |\operatorname{Lagr}(2n, \mathbb{F}_2)|/2^{2n-c}$ has average correlation at most $5\rho_{\text{avg}}$.*

Motivation: $k = 2$ isotropic pencils force any threshold above $4\rho_{\text{avg}}$ (their expected $2^{\dim(L\cap L')}$ ratio tends to 4 from below), while $k \geq 3$ pencils fall below the size threshold $|\operatorname{Lagr}|/2^{2n}$ for $n \geq 4$, and mixtures of pencils with random fill dilute quadratically. A proof would require classifying near-extremal subsets.

Theorem 6.6 (Conditional main SQ lower bound)**.** *Assume Conjecture 6.5. Any SQ algorithm distinguishing LSN (uniform L) from D_0 with probability $> 2/3$ requires $q \geq 2^{2n-O(1)}$ queries to* $\mathrm{VSTAT}(1/(15\rho_{\text{avg}}))$*.*

Proof. Under Conjecture 6.5 with $c = 0$, every subset of size at least $|\operatorname{Lagr}(2n, \mathbb{F}_2)|/2^{2n}$ has average correlation at most $5\rho_{\text{avg}}$. Hence $\mathrm{SDA}(B(\{D_L\}, D_0), 5\rho_{\text{avg}}) \geq 2^{2n}$. Applying Theorem 2.3 with $\alpha = 2/3$ and $\gamma = 5\rho_{\text{avg}}$ yields $q \geq (4/3 - 1) \cdot 2^{2n} = 2^{2n-O(1)}$ queries to $\mathrm{VSTAT}(1/(15\rho_{\text{avg}}))$. □

Theorem 6.7 (Adaptive SQ)**.** *Theorem 6.6 holds for randomized adaptive SQ algorithms.*

Proof. Theorem 2.3 applies to randomized adaptive SQ algorithms [FGR+17, Theorem 3.7], so the bound is inherited directly. □

6.6 Adaptive Linear SQ is Blocked

Theorem 6.8 (Adaptive linear SQ barrier)**.** *For any real-linear query $q(a, b) = \langle w, a\rangle + c \cdot b$, the expectation $\mathbb{E}_{D_L}[q]$ is independent of L. For any F_2-linear query $q(a, b) = \langle w, a\rangle \oplus c \cdot b$, the maximum distinguishing advantage against D_0 is $(1 - 2p)\, 2^{-n}$; at $p = 1/4$ this is 2^{-n-1}.*

Proof. For real-linear queries, direct computation gives $\mathbb{E}_a[\langle w, a\rangle] = 0$ and $\mathbb{E}_{D_L}[b] = p + 2^{-n}(1 - 2p)$, which is independent of L. For an F_2-linear query $q(a, b) = \langle w, a\rangle \oplus c \cdot b$: if $c = 0$ and $w \neq 0$ then $\mathbb{E}_{D_L}[q] = 1/2$ for every L (the constant query $w = 0, c = 0$ is trivially L-independent); if $c = 1$ and $w = 0$, then $q = b$ and the advantage over D_0 is exactly $(1 - 2p)2^{-n}$. If $c = 1$ and $w \neq 0$, write $q = b \oplus \langle w, a\rangle$. Under D_L we have

$$\mathbb{E}_{D_L}[q] = \frac{1}{2} + (1 - 2p)\big(2^{-n} - 2\mathbb{E}_a[\langle w, a\rangle \cdot \mathbf{1}_L(a)]\big).$$

If $w \in L^\perp$ then $\mathbb{E}_a[\langle w, a\rangle \cdot \mathbf{1}_L(a)] = 0$ and the advantage is $(1 - 2p)2^{-n}$. If $w \notin L^\perp$ then $\mathbb{E}_a[\langle w, a\rangle \cdot \mathbf{1}_L(a)] = 2^{-n-1}$ and the expectation collapses to $1/2$, giving advantage 0. Thus no F_2-linear query has advantage larger than $(1 - 2p)2^{-n}$. □

7 Quantum Analysis

7.1 Grover Search

The secret space has size $|\operatorname{Lagr}(2n, \mathbb{F}_2)| = 2^{n^2/2+O(n)}$. A Grover search over this space requires $2^{n^2/4+O(n)}$ iterations, each costing at least $\Omega(mn)$ to evaluate an LSN sample set. The total cost $2^{n^2/4+O(n)}$ dominates all classical bounds for $n \geq 4$, so Grover is not a practical threat.

7.2 Quantum BKW

A quantum version of BKW can speed up the inner Walsh–Hadamard transform and the search for low-weight parity checks, yielding complexity $2^{O(n^2/\log n)}$ [Kir11]. Since $n^2/\log n \gg 2n$ for all $n \geq 4$, this exceeds the SQ lower bound and is irrelevant for security parameter selection.

7.3 Quantum SQ

Because LSN samples are classical bits, any quantum algorithm must either access them through classical queries or perform a quantum walk over the sample space. A Grover search over the $2^{n^2/2}$-sized secret space still requires $2^{n^2/4}$ iterations (Section 7.1), far above the security parameter. Thus the exponential SQ lower bound provides evidence (not proof) that no efficient quantum algorithm solves LSN.

Conjecture 7.1 (Quantum LSN). *Any quantum algorithm solving $LSN_{n,m,p}$ with constant p requires $\Omega(2^n)$ queries.*

7.4 Dihedral Hidden Subgroup

The Lagrangian Grassmannian is not a group; it lacks a transitive abelian structure. Consequently, no reduction to the dihedral hidden-subgroup problem or its generalizations is known.

7.5 What Is and Is Not Covered

We summarize the quantum attack landscape honestly.

Blocked vectors. The following quantum approaches have been analyzed and ruled out for LSN at constant noise:

- **Hidden Subgroup Problem.** The Lagrangian Grassmannian lacks a group structure with coset decomposition; no reduction to DHSP or abelian HSP is known.
- **Grover search.** Complexity $2^{n^2/2+O(n)}$ exceeds all classical bounds.
- **Quantum BKW.** Complexity $2^{O(n^2/\log n)}$ is super-exponential in the security parameter.

Not covered. We do *not* claim a quantum lower bound for LSN. Quantum hardness is treated as a **conjecture** (Conjecture 7.1), analogous to the status of LWE and LPN. A rigorous quantum query lower bound would require new techniques beyond the SQ model and is left as future work.

8 Cryptographic Primitives

8.1 LSN-SNARK

We construct a preprocessing SNARK for the relation "$x \in L$" given a committed Lagrangian L. The construction exploits the *Siegel-chart* (graph-Lagrangian) representation to reduce the R1CS circuit from $O(n^3)$ to $O(n^2)$ constraints.

Siegel-chart representation. Fix a public symplectic matrix $S \in \mathrm{Sp}(2n, \mathbb{F}_2)$ (e.g., generated from a standard seed). Every "generic" Lagrangian can be written as

$$L_M = S \cdot \{(x, Mx) : x \in \mathbb{F}_2^n\},$$

where $M \in \mathrm{Sym}(n, \mathbb{F}_2)$ is a symmetric matrix. The set of graph Lagrangians has cardinality $|\mathrm{Sym}(n)| = 2^{n(n+1)/2}$, which differs from the full Lagrangian family $|\,\mathrm{Lagr}\,| = \prod_{i=1}^{n}(2^i + 1)$ by only a constant factor (≈ 2.38). Consequently, restricting to graph Lagrangians does not affect asymptotic security; the brute-force complexity remains $2^{n^2/2+O(n)}$ and the SQ lower bounds (Theorems 6.4 and 6.6) apply asymptotically unchanged because correlation structure is preserved under symplectic change of coordinates.

Binding M to the public key. The LSN-KEM public key (section 8.2) contains m noisy samples (x_i, y_i) generated from the secret Lagrangian L_M. To prove knowledge of M in a SNARK, the prover must bind their witness to the public key. The public key is therefore augmented with a commitment $c = \mathsf{Hash}(M)$, where Hash is a SNARK-friendly hash function (e.g., Poseidon). This introduces preimage-resistance of Hash as a second assumption; we size Hash (≥ 256-bit output, Poseidon) so that quantum preimage search (2^{128}) is at least the LSN level, making LSN the bottleneck. The SNARK circuit then verifies two things: (1) commitment opening ($c = \mathsf{Hash}(M)$), and (2) membership ($x \in L_M$). Hashing the $n(n+1)/2$ bits of M with Poseidon adds only $O(n)$ constraints—packing the bits into ≈ 9 field elements for $n = 65$ costs roughly 300–500 R1CS constraints. The full circuit size is therefore $n^2 + O(n)$, still $O(n^2)$ and still orders of magnitude below ZK-lattice circuits.

Membership circuit. The core membership sub-circuit verifies $x \in L_M$ in two steps:

1. Apply the public inverse S^{-1} to x to obtain $(a, b) = S^{-1}x$. This is a linear operation over $\mathbb{F}_2$, hence costs **zero** R1CS constraints.
2. Check $b = Ma$. Each entry $b_i = \sum_{j=1}^{n} M_{ij}a_j$ is a dot product of length n over $\mathbb{F}_2$. Treating a as a witness vector yields n AND gates per row, for n^2 constraints total; the final sums are linear and absorbed into the R1CS output side. Symmetry of M is free: we encode only the upper-triangular $n(n+1)/2$ entries.

   Further optimization. If a is provided directly as a public input (derived from x by the verifier), each product $M_{ij}a_j$ becomes a selector—free when $a_j = 0$ and a copy when $a_j = 1$. The check $b_i = \sum_j M_{ij}a_j$ then collapses to a single linear constraint per row, giving n **constraints** over $\mathbb{F}_2$ R1CS, or $n(n+3)/2$ constraints over a prime field with boolean witnesses for M.

Table 2 compares LSN-SNARK with other ZK-PQC approaches.

Table 2: R1CS constraints for LSN membership vs. ZK-PQC full primitives.

Scheme	Constraints ($n = 65$)	Asymptotic
LSN-SNARK membership	$\approx$ **4,225**	$\mathbf{O(n^2)}$
LSN-SNARK full (+ hash)	$\approx$ **4,500–4,700**	$\mathbf{O(n^2)}$
SHA-256 (per block)	$\approx$ 25,000	$O(1)$
AES-128	$\approx$ 16,000	$O(1)$
ZK-Kyber (full KEM verif.)	millions	$O(n \log n)$
ZK-SPHINCS+ (full sig. verif.)	millions	$O(\lambda \log \lambda)$

Witness structure. The witness $\mathbf{w}$ consists of:

- $n(n+1)/2$ bits for the upper-triangular entries of $M \in \mathrm{Sym}(n, \mathbb{F}_2)$;
- n bits for the vector a (the first half of $S^{-1}x$, computed as a linear function of x but included in the witness for the dot-product checks);
- auxiliary opening data for the hash commitment (omitted for deterministic hashes, negligible otherwise).

Total witness size: $n(n+1)/2 + O(n) = O(n^2)$ bits.

Verified counts. The n^2 prediction is exact for the membership sub-circuit when a is a witness variable: $n = 8$ yields 64 constraints, $n = 41$ (80-bit) yields 1,681, and $n = 65$ (128-bit) yields 4,225. With public a, the optimized membership counts are 8, 41, and 65 respectively over $\mathbb{F}_2$. The *full* circuit (membership + Poseidon commitment opening) at $n = 65$ is $\approx$ 4,500–4,700 constraints depending on hash parameterization.

Why this matters. The membership relation $x \in L_M$ thus has an unusually compact ZK circuit—$\approx$ 4,225 R1CS constraints at $n = 65$ (or 65 with public a), small relative to hashing one SHA-256 block ($\approx$ 25,000). The full primitive (membership + commitment opening) stays below $\approx$ 5,000 constraints—about 5× smaller than a single SHA-256 block ($\approx$ 25,000) and roughly 1,000× smaller than published ZK-Kyber circuits (millions). **Scope.** The table compares *core-relation* circuits; full signature/KEM verification includes additional binding overhead that is comparable across schemes. Our claim is not a record for the full signature, but that the *membership* sub-circuit—the inner relation being proved—is exceptionally compact. That compactness helps building blocks such as *verifiable PQ encryption* and *recursive SNARK aggregation*, where many membership checks dominate the cost.

Batch verification. The compactness enables dramatic amortization when proving membership for k distinct elements $x^{(1)}, \ldots, x^{(k)}$ under the same secret M. The commitment $c = \mathsf{Hash}(M)$ is verified **once**; each additional membership check adds only n^2 constraints (or n with public a). For batch size k:

$$\text{Total constraints} = \underbrace{O(n)}_{\text{hash opening}} + k \cdot \underbrace{O(n^2)}_{\text{membership}} .$$

In contrast, ZK-Kyber verification requires independent NTT evaluations per proof, offering no comparable amortization. Concrete example ($n = 65$, $k = 1000$): LSN uses $\approx 500+1000\cdot 4{,}225 \approx 4.2\times10^6$ constraints, whereas ZK-Kyber at $\approx 5\times10^6$ constraints per proof would need $\approx 5\times10^9$—a 1,000× gap.

8.2 LSN-KEM

8.2.1 Why concatenated codes

Direct polar coding for LSN at $p = 1/4$ fails. The Bhattacharyya parameter of BSC(p) is $Z_0 = 2\sqrt{p(1-p)} = 0.866$, too close to 1 for the finite-length polar code $N = 2048$ to achieve any useful reliability. We therefore concatenate:

- an inner repetition code of length r, reducing the effective channel to BSC(p') where

$$p' = \sum_{k=\lceil r/2 \rceil+1}^{r} \binom{r}{k} p^k (1-p)^{r-k};$$

- an outer polar code of length $N = 2048$ and dimension $K = 256$ over BSC(p'), decoded with SCL of list size $L = 8$.

Table 3 gives the parameters.

Table 3: LSN-KEM parameters for concatenated polar reconciliation.

Security	n	r	p'	SC bound	SCL estimate
80-bit	41	7	0.0706	2^{-81}	2^{-89}
128-bit	65	11	0.0343	2^{-149}	2^{-157}

8.2.2 Construction

We use the random-oracle model (ROM) for the security proof.

$\mathsf{KeyGen}(1^\lambda)$**:** Let $S \in \mathrm{Sp}(2n, \mathbb{F}_2)$ be a public symplectic matrix. Choose a symmetric matrix $M \leftarrow \mathrm{Sym}(n, \mathbb{F}_2)$ uniformly and define $L = S \cdot \{(x, Mx) : x \in \mathbb{F}_2^n\}$. Choose $\mathsf{seed} \leftarrow \{0,1\}^\lambda$. For $i = 1, \ldots, m$ (where $m = N \cdot r$), compute $x_i = \mathsf{PRG}(\mathsf{seed}, i)$ and $y_i = \mathbf{1}_L(x_i) \oplus e_i$. Output $\mathsf{pk} = (\mathsf{seed}, y_1, \ldots, y_m)$ and $\mathsf{sk} = M$.

Security remark. Restricting to graph Lagrangians reduces the key-space size by a constant factor $|\mathrm{Sym}(n)|/|\mathrm{Lagr}| \to \prod_{i \geq 1}(1 + 2^{-i})^{-1} \approx 1/2.38$, which does not affect the asymptotic security parameter. The SQ lower bounds (Theorems 6.4 and 6.6) and the empirical brute-force barriers (Section 5) apply asymptotically unchanged because correlation structure is preserved under symplectic change of coordinates. As a bonus, the secret key is more compact: M requires only $n(n+1)/2$ bits versus $2n^2$ bits for a general basis representation.

Encaps: Choose randomness $\rho = (s, u)$ with $s \leftarrow \{0,1\}^\lambda$ and $u \leftarrow \{0,1\}^K$. Compute a Fisher–Yates permutation $\pi = \mathsf{Permutation}(\mathsf{PRG}(s))$ over $[m]$, and let $\mathsf{indices} = \pi[: Nr]$. Group the selected indices into N blocks of length r and compute $v_j = \mathsf{majority}(y \text{ in block } j)$ for each block. Compute $c = \mathsf{PolarEncode}(u)$, set $\mathsf{ct} = (s, v \oplus c)$, and $K = \mathsf{Hash}(s, u)$.

Decaps: Recompute π and the blocks. For each block compute $v'_j = \mathsf{majority}(\mathbf{1}_L(x))$ using the secret key. Because the x's are derived from the same public seed, the v'_j are deterministic; they need not equal the noisy v_j sent by the encapsulator, but the difference $v_j \oplus v'_j$ is exactly the effective noise seen by the polar decoder. Compute $c' = \mathsf{syn} \oplus v'$, decode $u' = \mathsf{PolarSCLDecode}(c')$, and output $K = \mathsf{Hash}(s, u')$ (or $\perp$ if decoding fails).

The permutation-based index sampling guarantees distinct indices, eliminating birthday collisions. Public-key size is $\lambda + Nr$ bits; ciphertext size is $\lambda + N$ bits.

Implementation note. A small-scale Python prototype of the SCL decoder exhibited a bug (BLER $= 1.0$ for $N = 128, 256$), contradicting the conservative Bhattacharyya bound and indicating an implementation issue rather than a parameter problem. Subsequent validation with the verified `komm` SC decoder [komm] revealed a second bug in the Bhattacharyya parameter indexing: the original code concatenated all bad channels before all good channels, whereas `komm`'s natural-order polar code requires interleaving ($z_{2i} = 2z_i - z_i^2$, $z_{2i+1} = z_i^2$). After correcting the frozen-set construction, all tested configurations yielded BLER $= 0$: at $p' = 0.0706$ for $N = 128, 256, 512$ with $K = N/8$, and at $p' = 0.0343$ for all tested lengths. The previous $N = 256$ anomaly (BLER $= 0.115$) was entirely due to the indexing mismatch, not incomplete polarization. These short-length, 200-trial simulations are *preliminary*: they certify only BLER $\lesssim 1/200$, not the design point 2^{-80}, and they do not test the design length $N = 2048$. The conservative Arıkan–Bhattacharyya bound (2^{-80} at $r = 7$, 2^{-148} at $r = 11$) remains the design basis; full $N = 2048$ validation is deferred to the production Rust implementation.

8.2.3 Correctness

Theorem 8.1 (Correctness)**.** *For $r = 7$, $\Pr[K \neq K'] < 2^{-80}$; for $r = 11$, $\Pr[K \neq K'] < 2^{-128}$.*

Proof. Inner majority vote converts BSC(1/4) into BSC(p'). The outer polar code Bhattacharyya bound gives $P_e \leq \frac{1}{2}\sum_{i \in \mathcal{I}} Z(W_i)$, where $\mathcal{I}$ is the information set. Numerical evaluation of the bound yields the claimed values; SCL decoding with $L = 8$ improves them by a small constant factor. □

8.2.4 IND-CPA Security

Theorem 8.2 (IND-CPA)**.** *LSN-KEM is IND-CPA secure in the random-oracle model under the LSN hardness assumption.*

Proof sketch. The proof uses three game hops.

1. **Game 0 → Game 1:** Replace PRG with a truly random function. Indistinguishability follows from PRG security.

2. **Game 1 → Game 2:** Replace the public-key labels $(y_1, \ldots, y_m)$ with uniform random bits. We construct a direct reduction $\mathcal{B}$ from KEM IND-CPA to LSN: $\mathcal{B}$ receives an LSN challenge (x_i, y_i^*), programs seed so that $\mathsf{PRG}(\mathsf{seed}, i) = x_i$, and sets $\mathsf{pk} = (\mathsf{seed}, y_1^*, \ldots, y_m^*)$. The KEM challenger computes ct^* using all $m = Nr$ samples. If the LSN challenge is real, the adversary sees Game 1; if random, Game 2. Any advantage ε in distinguishing the games translates to advantage ε against LSN. There is no $1/r$ security loss because the reduction uses the LSN challenge directly.

3. **Game 2 → Game 3:** In Game 2 the vector v is uniform and independent of the polar codeword $c = \mathsf{PolarEncode}(u)$. Hence $\mathsf{syn} = v \oplus c$ is uniform (one-time pad), and $K = \mathsf{Hash}(s, u)$ is uniform and independent of ct^* by the random-oracle property. □

8.2.5 Sizes and Comparison

Table 4 compares LSN-KEM with selected post-quantum KEMs.

Table 4: KEM size comparison.

Scheme	$\lvert\mathsf{pk}\rvert$	$\lvert\mathsf{ct}\rvert$	Assumption	Failure
Kyber-512	800 B	768 B	Module-LWE	2^{-164}
HQC-128	2.25 KB	4.50 KB	Syndrome Dec.	2^{-128}
LSN-80	1.78 KB	288 B	LSN	$< 2^{-89}$
LSN-128	2.78 KB	288 B	LSN	$< 2^{-157}$

8.2.6 CCA Security

We obtain IND-CCA security via the Fujisaki–Okamoto (FO) transform [FO99, HHK17] with implicit rejection. Let $\mathsf{BaseKEM} = (\mathsf{KeyGen}, \mathsf{Encaps}, \mathsf{Decaps})$ be the LSN-KEM of section 8.2.2; recall that Encaps is deterministic once the random pair (s, u) is fixed. Define a hash function $H : \{0,1\}^* \to \{0,1\}^\lambda$ (modelled as a random oracle).

$\mathsf{CCA\text{-}KeyGen}(1^\lambda)$: $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$. Output $\mathsf{sk}' = (\mathsf{sk}, d)$ where $d \leftarrow \{0,1\}^\lambda$ is an implicit-rejection secret.

$\mathsf{CCA\text{-}Encaps}(\mathsf{pk})$: Choose $s \leftarrow \{0,1\}^\lambda$ and $u \leftarrow \{0,1\}^K$, compute $(c_0, k_0) \leftarrow \mathsf{Encaps}(\mathsf{pk}; (s, u))$, set $c = c_0$ and $K = H(s, u, c_0)$. Output (c, K).

$\mathsf{CCA\text{-}Decaps}(\mathsf{sk}', c)$: Parse $c = (s, \mathsf{syn})$. Recompute the blocks from s and $v_j' = \mathsf{majority}(\mathbf{1}_L(x))$ as in base decaps. Compute $c' = \mathsf{syn} \oplus v'$ and decode $u' = \mathsf{PolarSCLDecode}(c')$. If $u' = \perp$, output $K = H(d, c)$ *(implicit rejection)*. Re-encapsulate $(c_0', k_0') \leftarrow \mathsf{Encaps}(\mathsf{pk}; (s, u'))$. If $c_0' = c$, output $K = H(s, u', c)$; otherwise output $K = H(d, c)$ *(implicit rejection)*.

Theorem 8.3 (IND-CCA)**.** *If* $\mathsf{BaseKEM}$ *is δ-correct with $\delta \le 2^{-80}$ and IND-CPA secure, then* $\mathsf{CCA\text{-}KEM}$ *is IND-CCA secure in the random-oracle model.*

Proof sketch. The FO transform preserves IND-CPA security and adds CCA robustness via the re-encapsulation check. Implicit rejection ensures that decapsulation-failure events are indistinguishable from pseudorandom outputs; we inherit the multi-challenge ROM bound of [HHK17]. CCA security implies non-malleability, so tampered ciphertexts are rejected by the re-encapsulation check. The concatenated code affects only the correctness error δ; since $\delta \leq 2^{-80}$ (Theorem 8.1), the standard FO analysis [HHK17] applies unchanged. $\square$

8.2.7 Multi-User Security

In the multi-user setting, N parties generate independent key pairs $(\mathsf{pk}_i, \mathsf{sk}_i)$ with independent secrets $M_i \leftarrow \mathrm{Sym}(n, \mathbb{F}_2)$. The adversary receives all N public keys and may query decapsulation oracles for any user.

Reduction to single-user security. All users share only the public symplectic matrix S; every other parameter (seed_i, noise $e_{i,j}$, permutation π_i) is independent. Because the M_i are independent and the PRG seeds are independent, the N instances are *statistically independent* conditioned on S. By a standard hybrid argument over the N users, any adversary $\mathcal{A}$ against N-user IND-CCA security with advantage ε_N yields an adversary against single-user IND-CCA security with advantage $\varepsilon_1 \geq \varepsilon_N/N$.

Key-reuse analysis. A more subtle concern is repeated encapsulation to the *same* user. Over q encapsulations the adversary sees $q \cdot m = qNr$ samples (x_j, y_j) from the same secret M. The brute-force decoder must still search $2^{n(n+1)/2}$ symmetric matrices, and the conditional SQ lower bound (Theorem 6.6) requires $q_{\mathsf{sq}} \geq 2^{2n-O(1)}$ queries. Translating sample count to query count via $q_{\mathsf{sq}} = qNr$, the number of encapsulations needed for non-negligible advantage satisfies

$$q \gtrsim \frac{2^{2n-O(1)}}{Nr}.$$

(We treat each public sample as at most one SQ query; the R2a experiment supports a 2^{2n} *sample* threshold.) For $n = 65$, $N = 2048$, $r = 11$ we have $Nr \approx 2^{14.5}$, giving $q \gtrsim 2^{114}$—far beyond any realistic threat. We therefore treat multi-user and key-reuse security as *straightforward* consequences of the single-user bound, with no special vulnerability introduced by the LSN structure.

8.3 Implementation Security

Constant-time by construction. All core LSN operations are native $\mathbb{F}_2$ operations—XOR, AND, and bit-permute—which are straightforward to implement in constant time on modern CPUs. There is no modular reduction with an odd modulus, no discrete Gaussian sampling, no rejection sampling, and no variable-time branching on secret data. This makes a constant-time implementation *straightforward* rather than requiring careful engineering as in lattice schemes.

Table 5: Implementation-security comparison: Kyber vs. LSN.

Operation	Kyber	LSN
Modular reduction	Variable-time ($q = 3329$)	**None** ($\mathbb{F}_2$)
NTT butterfly	Twiddle-dependent memory access	**Fixed** XOR pattern
Noise sampling	Discrete Gaussian (rejection)	**Uniform** bits
Decoder	Noisy polynomial rounding	**Majority** + fixed-tree SC
Secret-key format	$2n^2$ bits (general basis)	$\mathbf{n(n+1)/2}$ bits (symmetric M)

Side-channel surface. The only non-linear operation on secret data is the majority vote in the inner repetition decoder. A popcount implementation is data-independent on all modern architectures (ARM NEON, AVX-512 VPOPCNTDQ). The outer polar SC decoder traverses a fixed butterfly tree of depth $\log_2 N$; the memory access pattern depends only on N, not on the secret M or the noise. Consequently, timing attacks on the decoder require exploiting micro-architectural effects in the popcount instruction itself—far narrower than the modular-reduction and Gaussian-sampling side channels of lattice KEMs.

Current status. A Python prototype validates parameters and algorithms. A production constant-time Rust implementation (including constant-time popcount, bit-sliced symplectic transforms, and a constant-time SC decoder) is planned for full $N = 2048$ validation and KAT vector generation.

9 Reduction Barriers and the Standard-Model Gap

Table 6 summarizes the status of natural reductions.

Table 6: Reduction landscape for LSN.

Class	Status	Reason
Linear (real)	BLOCKED	real-linear: $\mathbb{E}[q]$ L-independent; F_2-linear: $\leq (1-2p)2^{-n}$
Polynomial degree $< n$	BLOCKED	Error $\geq 2^{-n}$
Linear sample-red., $m = \Theta(n)$	RULED OUT	LPQR26 (external)
Linear sample-red., $m = \omega(n)$	OPEN	LPQR26 bound insufficient (their caveat; extension expected)
LPN($k = \Theta(n^2)$)	VACUOUS	Proposition 9.1 + BKW cost $2^{\Omega(n^2/\log n)}$
Multi-sample or beyond-poly	OPEN	No proof or construction known

Lu *et al.* [LPQR26] prove that linear reductions cannot reduce sympLPN to LPN (their §2.4 and Appendix D): for any *fixed* $B \in \mathbb{F}_2^{m\times 2n}$ the matrix BA has entropy at most $(1-d)mn$ for a constant d, so randomizing the public matrix forces a high-entropy B, which drives the output error weight past the Shannon-converse threshold. Per their Appendix D, the proof targets the primary cryptographic regime $m = \Theta(n)$ (output rows $m = (1+\epsilon)n$); for $m = \omega(n)$ it shows error weight larger than $1/2 - \delta$ for any constant δ, which they note is not sufficient to fully rule out decodability, though they state they expect the barrier to extend to all $m = \mathrm{poly}(n)$. Our Theorem 6.8 gives a complementary single-query barrier of $(1-2p)2^{-n}$ at any noise rate.

A sharper, unconditional barrier comes from an information-theoretic floor:

Proposition 9.1 (Entropy floor)**.** *Any reduction from* $\mathrm{LSN}(n)$ *to* $\mathrm{LPN}(k)$ *that preserves the secret with recovery probability* $\alpha(n)$ *must satisfy*

$$k \;\geq\; \alpha(n)\, H(L) - O(1) \;=\; \alpha(n)\,\Theta(n^2).$$

In particular, constant success probability requires $k = \Omega(n^2)$*, and success probability* $1 - o(1)$ *requires* $k \geq H(L) - o(n^2)$*.*

Proof. The LSN secret space is $\mathrm{Lagr}(2n, \mathbb{F}_2)$ with $|\,\mathrm{Lagr}\,| = \prod_{i=1}^{n}(2^i + 1)$. Hence $H(L) = \log_2 |\,\mathrm{Lagr}\,| = n(n+1)/2 + \Theta(1)$, where the additive constant is $\log_2 \prod_{i=1}^{\infty}(1 + 2^{-i}) \approx 1.25$. If a reduction outputs an $\mathrm{LPN}(k)$ instance from which L is recovered with probability α, Fano's inequality gives $k \geq \alpha H(L) - H_2(1-\alpha) = \alpha H(L) - O(1)$. □

Concrete-security win-win. For near-perfect recovery of L (i.e., $\alpha = 1 - o(1)$ in Proposition 9.1), the entropy floor forces $k \geq H(L) - o(n^2)$. For the parameter sets in Table 1 this gives $k \geq 863$ (80-bit), $k \geq 2147$ (128-bit), $k \geq 4755$ (192-bit), or $k \geq 8387$ (256-bit). The best known attack on LPN(k) (BKW [BKW03]) requires time $2^{\Omega(k/\log k)}$. At these dimensions this yields $2^{88.5}$, $2^{194.0}$, $2^{389.3}$, and $2^{643.5}$ respectively—all exceeding the target security levels. We stress that this is a concrete-security observation, not an asymptotic impossibility: BKW on LPN($\Theta(n^2)$) runs in $2^{\Theta(n^2/\log n)}$, which is asymptotically smaller than brute-force on LSN ($2^{\Theta(n^2)}$). For the parameter range of interest, however, the entropy floor lands BKW well above the security budget.

Single-sample adaptivity does not increase degree. A natural question is whether *adaptive* degree-D feature-map reductions circumvent the static degree barrier. For the restricted model in which the reduction is *single-sample*—it may choose each ϕ_j adaptively, but applies it only to the raw samples (a, b), with the choice of ϕ_j made as a function of previous answers—the answer is negative: in each round the feature map $\phi_j(a, b)$ has degree at most D, and the effective degree of the entire transcript is bounded by D. When $D < n$, the Reed–Muller obstruction of **??** still applies: correlation with $\mathbf{1}_L$ is at most 2^{-n}, requiring exponentially many samples. Thus single-sample adaptive degree-D feature-map reductions are blocked for every $D < n$, independent of the number of rounds.

The remaining open question is whether a *multi-sample* or *genuinely non-linear* adaptive reduction exists. Such a reduction could apply non-linear functions to *fresh* samples generated from previous outputs, or could use adaptively chosen non-linear queries in a way that raises the algebraic degree of the transcript. We do not claim impossibility for this richer model. We do note a win-win structure: a reduction in this richer model would likely improve random self-reductions for LPN itself, making it valuable even if it collapses LSN to the 6.5th family.

The symplectic structure is reduction-inert. A sharper observation comes from the structure of the LSN samples themselves. The deterministic quadratic constraint $S_A = 0$ (the Ω-Gram of the public matrix columns) is *public* and *x-free*: it depends only on the public matrix A, not on the secret x or the noise. Consequently, $S_A = 0$ adds no x-information to an attacker and provides no usable handle for a reductionist trying to embed LPN into LSN. This mirrors the accepted status of Ring-LWE, which is regarded as a lattice-family assumption by *source* (ring structure native to the problem) rather than by reduction to plain LWE. We formalize this as:

Conjecture 9.2 (Source novelty)**.** *The following three invariants of LSN have no analogue in standard LPN, and each is destroyed by every natural reduction class that has been proposed:*

1. ***Symplectic-Fourier self-duality.*** *The indicator $\mathbf{1}_L$ satisfies $\mathcal{F}_\Omega[\mathbf{1}_L] = 2^n \mathbf{1}_L$, where $\mathcal{F}_\Omega$ is the symplectic Fourier transform. A standard linear function $f(x) = \langle s, x\rangle$ has no such fixed-point property; its Fourier transform is a Dirac mass, not a function.*

2. ***x-free quadratic constraint.*** *The public matrix A in LSN satisfies $S_A = 0$ (the Ω-Gram of columns is identically zero). This is a deterministic, public, quadratic constraint on the* public parameters *alone. LPN has no analogous public structure linking its samples.*

3. ***Stabilizer degeneracy.*** *Every LSN secret L is a stabilizer state with stabilizer weight $\Omega(n)$. The secret space is the Lagrangian Grassmannian, not a vector space. LPN secrets are bare linear functions.*

Each invariant is preserved by the symplectic group $\mathrm{Sp}(2n)$ *and destroyed by any of the blocked reductions: linear maps lose the self-duality (Lu et al. [LPQR26]), degree-$D < n$ polynomial maps cannot represent $\mathbf{1}_L$ (Reed–Muller distance), and entropy-compressing maps discard the stabilizer structure (Proposition 9.1).*

Conjecture 9.2 is falsifiable: a single explicit reduction sympLPN $\to$ LPN that preserves any of the three invariants would refute the conjecture in its current form. It is not decidable by reduction analysis alone; it rests on the absence of such a reduction and the structural barriers documented above.

The scrambling barrier at constant noise. The LPQR26 linear barrier applies at any noise rate, and at our constant noise $p = 1/4$ its error-amplification step is in fact *stronger*: a randomizing row of weight w leaves per-bit bias $(1-2p)^w = 2^{-w}$ (piling-up), versus $\approx 1 - 2wp$ in the low-noise regime. The residual gaps are therefore not about noise: (i) linear reductions consuming $m = \omega(n)$ samples, where the LPQR bound does not fully rule out decodability (their Appendix D); and (ii) non-linear scramblers. For (ii) an algebraic obstruction remains: $S_A = 0$ is a deterministic quadratic relation among the public vectors (our observation, distinct from the LPQR entropy argument). Direct inspection of the $n = 3$ case shows that simple scramblers fall into one of three classes: a linear scrambler *transports* the quadratic relation ($S'_{A'} = 0$ with respect to the conjugated form $M^{-\top}\Omega M^{-1}$, so it remains detectable), a non-linear scrambler destroys the linear label structure required by LPN, and a random scrambler destroys the secret entirely. None of these simple classes simultaneously destroys $S_A = 0$ and creates a fresh LPN secret-bit structure. The residual question — whether a sufficiently complex non-linear scrambler exists — is the same as Open Problem 2.

10 Open Problems

We state precisely the problems whose resolution would close the standard-model gap.

1. **Statistical dimension under $S_A = 0$ [LPQR26].** The full SQ proof is blocked by the deterministic quadratic symplectic relation $S_A = 0$ (the Ω-Gram of columns). Prove that conditioning on $S_A = 0$ does not reduce the statistical dimension below $2^{\Omega(n)}$.

2. **LPN(low-noise) $\to$ LSN.** In the cryptographically relevant regime $p = O(1/\sqrt{n})$, the reduction of [KLP+25] is vacuous. Does a non-vacuous reduction exist, or can its non-existence be proved?

3. **LWE $\to$ LSN.** A reduction from LWE (or worst-case lattice problems) to LSN would provide the gold-standard foundation currently missing.

4. **Quantum lower bound.** Prove Conjecture 7.1: any quantum algorithm solving LSN requires $\Omega(2^n)$ queries.

5. **Optimal polar-code rate.** Maximize the rate of the outer polar code while keeping decryption failure below 2^{-128}.

6. **Pencil-extremality conjecture.** Prove Conjecture 6.5: that isotropic pencils ($k \leq 2$) are extremal at scale $|\operatorname{Lagr}(2n, \mathbb{F}_2)|/2^{2n}$, giving average correlation $\leq 5\rho_{\text{avg}}$ for every subset of that size. This would upgrade Theorem 6.6 from a conditional to an unconditional worst-case bound.

7. **Sample freshness for rerandomized LSN.** Exact secret rerandomization is possible: applying a public symplectic matrix $S \in \operatorname{Sp}(2n)$ to the public part of each sample maps the secret L to $S \cdot L$ while preserving labels. Does LSN admit a public transformation that yields *fresh* (statistically independent) samples for the rerandomized secret? A positive answer would yield a tight multi-user security bound and improve the reduction in Section 8.2.7. A negative answer would explain why the hybrid argument loses a factor N.

10.1 Honest Limitations of the Present Work

We flag five practical gaps that a reader should weigh against the theoretical claims above.

1. **Empirical gap at $N = 2048$.** The concatenated code is designed via the Bhattacharyya bound, which guarantees $P_e \leq 2^{-80}$ for the target block length $N = 2048$. We have validated the SC decoder empirically for $N \in \{128, 256, 512\}$ (all BLER = 0 in 200 trials), but a direct Monte-Carlo run at $N = 2048$ is computationally expensive and has not been completed. The correctness claim for $N = 2048$ therefore rests on the Bhattacharyya bound and the smaller-scale validation, not on a direct measurement.

2. **No constant-time reference implementation.** While section 8.3 argues that LSN-KEM is constant-time "by construction," this property has not been verified on a concrete, audited implementation. A production-grade Rust reference implementation with known-answer test (KAT) vectors is planned but not yet available. Physical fault-injection resistance (e.g., bit-flip attacks on the majority vote) is likewise deferred to the audited implementation.

3. **Full-protocol SNARK is future work.** Section 8.1 optimizes only the *membership* sub-circuit (n^2 constraints). A complete zero-knowledge proof of correct encapsulation—including the polar encoder, the majority-vote decoder, and the random permutation—remains open. The comparison in Table 2 should be read as a lower bound on what a full SNARK would cost.

4. **Loose multi-user reduction.** Section 8.2.7 uses a hybrid argument losing a factor N. A tight reduction would require sample freshness for a rerandomized secret. Exact secret rerandomization *is* possible—applying a public symplectic matrix $S \in \mathrm{Sp}(2n)$ to the public part of each sample maps L to $S \cdot L$ while preserving labels—but *sample freshness* is blocked: because $\mathbf{1}_L$ is non-linear, one cannot generate independent fresh samples for the new secret from the old public samples alone without knowing L. In practice the margin is comfortable ($q \gtrsim 2^{114}$), but the bound is not tight.

5. **Practical optimizations deferred.** The canonical noise rate $p = 1/4$ is fixed; whether LSN remains viable at other rates is unanalyzed. The inner repetition code uses a majority-vote decoder for constant-time implementability, but MAP decoding would achieve a lower effective noise. No public-key compression analogous to structured LPN is known. These are engineering tuning degrees of freedom, not structural vulnerabilities.

Threshold cryptography. The LSN secret M admits a natural linear secret-sharing scheme over $\mathbb{F}_2$ ($M = \sum M_j$), and the public samples (x_i, y_i) provide a publicly verifiable consistency check because the linear relation $b = Ma = \sum_j M_j a$ is additively shareable. This enables distributed key generation where n parties jointly sample M without any single party learning it. Formalizing robustness against malicious minorities is future work.

Quantum ECC interoperability. Every LSN secret L is a stabilizer state ($k = 0$). Its minimum stabilizer weight is $\Omega(n)$ with high probability by the quantum Gilbert–Varshamov bound, so the secret carries natural quantum error-correcting structure. Whether this classical–quantum linkage yields concrete protocols (e.g., quantum-authenticated key exchange) is an open application direction.

A Full Proof of Pairwise Correlation

Here we give a self-contained derivation of Lemma 6.1. Let D_0 be the distribution on $\mathbb{F}_2^{2n} \times \mathbb{F}_2$ in which a is uniform and $b \sim \mathrm{Bernoulli}(p)$ independently. For a fixed Lagrangian L, the LSN distribution D_L has density

$$D_L(a,b) = \frac{1}{2^{2n}}\Big((1-p)\mathbf{1}_{b=\mathbf{1}_L(a)} + p\mathbf{1}_{b\neq\mathbf{1}_L(a)}\Big).$$

The likelihood ratio is

$$\frac{D_L(a,b)}{D_0(a,b)} = 1 + \mathbf{1}_L(a)\cdot\beta\cdot\Big(\mathbf{1}_{b=1} - \mathbf{1}_{b=0}\frac{p}{1-p}\Big),$$

where $\beta = (1-2p)/p$. Therefore

$$\ell_L(a,b) = \mathbf{1}_L(a)\cdot\beta\cdot\Big(\mathbf{1}_{b=1} - \mathbf{1}_{b=0}\frac{p}{1-p}\Big).$$

For two secrets L, L' with $j = \dim(L\cap L')$,

$$\begin{aligned}
\langle D_L, D_{L'}\rangle &= \mathbb{E}_{D_0}[\ell_L\ell_{L'}] \\
&= \mathbb{E}_a[\mathbf{1}_L(a)\mathbf{1}_{L'}(a)]\cdot\beta^2\mathbb{E}_b\Big[\Big(\mathbf{1}_{b=1} - \mathbf{1}_{b=0}\frac{p}{1-p}\Big)^2\Big] \\
&= 2^{j-2n}\cdot\beta^2\Big(p + (1-p)\frac{p^2}{(1-p)^2}\Big) \\
&= 2^{j-2n}\cdot\frac{(1-2p)^2}{p(1-p)}.
\end{aligned}$$

Setting $p = 1/4$ gives the coefficient $(1/2)^2/(1/4\cdot 3/4) = (1/4)/(3/16) = 4/3$ exactly.

B Q-Binomial Distance Distribution

The number of k-dimensional subspaces of $\mathbb{F}_2^n$ is the Gaussian binomial coefficient

$$\begin{bmatrix} n \\ k \end{bmatrix}_2 = \prod_{i=0}^{k-1}\frac{2^{n-i}-1}{2^{k-i}-1}.$$

To count pairs of Lagrangians with intersection dimension k, fix a k-dimensional isotropic subspace W. The number of Lagrangians containing W equals the number of Lagrangians in the symplectic quotient $W^\perp/W \cong \mathbb{F}_2^{2(n-k)}$, namely $\prod_{i=1}^{n-k}(2^i+1)$. Summing over all k-dimensional subspaces $W \subseteq L$ and using the identity $\prod_{i=1}^{n-k}(2^i+1) = 2^{(n-k)(n-k+1)/2}\prod_{i=1}^{n-k}(1+2^{-i})$, the exact probability is

$$\Pr[j = k] = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_2\cdot 2^{(n-k)(n-k+1)/2}}{\prod_{i=1}^{n}(2^i+1)}.$$

The moment generating function $C_n = \mathbb{E}[2^j]$ converges rapidly to 2 as $n\to\infty$.

C Reduction Barrier Summary

Linear. For any linear classifier $h(a) = \langle w, a\rangle + t$, the advantage

$$\mathrm{Adv}(h) = \big|\Pr_{D_L}[h(a)=1] - \Pr_{D_0}[h(a)=1]\big|$$

is zero because $\Pr_{D_L}[h(a)=1] = \Pr_{D_0}[h(a)=1]$.

Polynomial degree $< n$. The exact polynomial representation of $\mathbf{1}_L$ has degree exactly n and 2^n monomials. Any polynomial of degree $d < n$ cannot vanish on L^c, so its correlation with $\mathbf{1}_L$ is at most 2^{-n}.

Adaptive linear SQ. For real-linear queries $q(a,b) = \langle w, a\rangle + cb$ we have $\mathbb{E}_{D_L}[q] = c(p + 2^{-n}(1-2p))$, independent of L. For F_2-linear queries the maximum advantage is $(1-2p)2^{-n}$ (Theorem 6.8). In either case the advantage is exponentially small in n, so adaptive linear SQ algorithms require exponentially many samples.

D Extension to $\mathbb{F}_q$

All results in the main text are stated over $\mathbb{F}_2$. Here we sketch the extension to arbitrary finite fields $\mathbb{F}_q$, which widens the parameter space and allows a trade-off between block length n and field size q.

D.1 Symplectic Spaces over $\mathbb{F}_q$

Let q be a prime power and $V = \mathbb{F}_q^{2n}$ equipped with a non-degenerate alternating bilinear form $\Omega(x,y) = x^T J y$ where $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$. A subspace $L \subset V$ is *Lagrangian* if it is maximal isotropic; then $\dim L = n$ and $|\,\mathrm{Lagr}(2n, \mathbb{F}_q)| = \prod_{i=1}^{n}(q^i + 1)$.

For fixed L and $j = \dim(L \cap L')$, the number of Lagrangians L' with intersection dimension j is

$$N_j(n,q) = \begin{bmatrix} n \\ j \end{bmatrix}_q \cdot q^{(n-j)(n-j+1)/2},$$

where $\left[\begin{smallmatrix} n \\ j \end{smallmatrix}\right]_q$ is the Gaussian binomial coefficient. Summing over j recovers $|\,\mathrm{Lagr}(2n, \mathbb{F}_q)|$.

D.2 Exact Correlation over $\mathbb{F}_q$

The label remains binary ($b \in \mathbb{F}_2$) even though the ambient space is $\mathbb{F}_q^{2n}$, so the SQ machinery of Feldman *et al.* applies without modification. The likelihood ratio calculation is identical with 2 replaced by q:

Lemma D.1 ($\mathbb{F}_q$ pairwise correlation)**.** *For $L, L' \in \mathrm{Lagr}(2n, \mathbb{F}_q)$ with $j = \dim(L \cap L')$,*

$$\langle D_L, D_{L'}\rangle = \frac{(1-2p)^2}{p(1-p)} \cdot \frac{q^j}{q^{2n}}.$$

Lemma D.2 ($\mathbb{F}_q$ average correlation)**.** *Let $C_{n,q} = \mathbb{E}[q^j]$ computed via the q-binomial distribution. Then*

$$\rho_{\mathrm{avg}} = \frac{(1-2p)^2}{p(1-p)} \cdot C_{n,q} \cdot q^{-2n}.$$

Numerically, $C_{n,q} \to 2$ as $n \to \infty$ for every q (the limit is independent of the field size).

D.3 Security Parameters over $\mathbb{F}_q$

The SQ lower bound becomes $q_{\mathrm{queries}} \geq q^{2n-O(1)}$. One "bit" of security corresponds to $\log_2(q)$ field operations, so the effective bit-security is $2n\log_2(q)$. Table 7 gives the block lengths n needed for target security levels.

Table 7: LSN parameters over $\mathbb{F}_q$ for $p = 1/4$.

Security	$\mathbb{F}_2$: n	$\mathbb{F}_3$: n	$\mathbb{F}_5$: n	$\mathbb{F}_7$: n
80-bit	41	26	18	15
128-bit	65	41	28	24
192-bit	97	62	42	35
256-bit	129	82	56	46

Larger q reduces the required n but increases the cost of each sample (field arithmetic in $\mathbb{F}_q$ versus $\mathbb{F}_2$). The optimal choice depends on the target platform.

D.4 Open Questions

1. Extend to q-ary labels $y \in \mathbb{F}_q$ with noise over $\mathbb{F}_q$; this requires q-ary SQ lower bounds.
2. The proof holds for $q = p^k$ with $k > 1$ because symplectic theory is field-agnostic.
3. In characteristic 2, alternating is equivalent to symmetric; in odd characteristic they differ. The intersection distribution formula remains valid in all characteristics.

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